SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

<u>Task</u>

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

SOL; **Step 1: Identify the Subnet**

Given:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)

A **/26** subnet has:

- Block size = **64**
- Subnets:
  - 192.168.10.0 – 192.168.10.63
  - **192.168.10.64 – 192.168.10.127**
  - 192.168.10.128 – 192.168.10.191
  - 192.168.10.192 – 192.168.10.255

Configured IP 192.168.10.65 belongs to subnet 192.168.10.64/26, **not** 192.168.10.0/26.

Step 2: Network Details

Parameter	**Value**
Network Address	**192.168.10.64**
Broadcast Address	**192.168.10.127**
Valid Host Range	**192.168.10.65 – 192.168.10.126**

Step 3: Verify IP Address and Gateway

- **IP Address (192.168.10.65):** ✅ Valid host address.
- **Default Gateway (192.168.10.1):** ❌ Invalid.

Reason:

- The gateway **192.168.10.1** belongs to subnet **192.168.10.0/26**.
- The workstation is in subnet **192.168.10.64/26**.
- The IP address and gateway must be in the **same subnet**.

Step 4: Configuration Errors

1. ❌ Workstation IP is **not in the intended subnet (192.168.10.0/26)**.
2. ❌ Default gateway belongs to a different subnet.
3. ❌ Due to subnet mismatch, the workstation cannot communicate with devices in **192.168.10.0/26**.

Step 5: Optimized Configuration

To communicate with subnet **192.168.10.0/26**, assign:

Parameter	Correct Value
IP Address	**192.168.10.10** (or any host from 192.168.10.1–192.168.10.62 except gateway)
Subnet Mask	**255.255.255.192 (/26)**
Default Gateway	**192.168.10.1**

Step 6: Usable Host Addresses

For a **/26** subnet:

- Total addresses = **64**
- Network address = **1**
- Broadcast address = **1**

Usable Hosts = 64 − 2 = 62

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

IPv6 Configuration Debugging

Step 1: Expand Both IPv6 Addresses

Device A

- Compressed: **2001:db8::ff00:42:8329/64**
- Expanded: **2001:0db8:0000:0000:0000:ff00:0042:8329/64**

Device B

- Compressed: **2001:db8:0:0:1::1/64**

- Expanded: **2001:0db8:0000:0000:0001:0000:0000:0001/64**

Step 2: Determine the Network Prefix

A **/64** prefix means the **first 64 bits (first four blocks)** represent the network.

Device	Network Prefix
Device A	**2001:db8:0:0::/64**
Device B	**2001:db8:0:0::/64**

Step 3: Verify Whether Both Devices Belong to the Same Subnet

- **Device A Prefix:** 2001:db8:0:0::/64
- **Device B Prefix:** 2001:db8:0:0::/64

Answer: ✅ **Yes, both devices belong to the same subnet.**

Step 4: Identify the Addressing or Prefix Mismatch

There is **no IPv6 addressing or prefix mismatch** because:

- Both devices have the same **/64** prefix.
- Both are in the **2001:db8:0:0::/64** network.

If communication fails, the cause is likely one of the following:

- Incorrect default gateway
- Interface is down
- Duplicate IPv6 address
- Firewall blocking traffic
- Neighbor Discovery (NDP) issue
- Switch/router configuration problem

Step 5: Optimized IPv6 Configuration

Use addresses within the same subnet:

Device	IPv6 Address
Device A	**2001:db8:0:0::10/64**
Device B	**2001:db8:0:0::20/64**

Both devices remain in the **2001:db8:0:0::/64** subnet.

Step 6: Number of Available Host Addresses

For a **/64** IPv6 subnet:

- Host bits = **64**
- Total available host addresses = $\mathbf{2^{64}}$

$\mathbf{2^{64}}$ **= 18,446,744,073,709,551,616** host addresses.

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3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.
<u>Task</u>
Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

SOL:**Step 1: Identify the Subnets**

Subnet	**Network Address**	**Broadcast Address**	**Valid Host Range**	**Usable Hosts**
/26	192.168.1.0	192.168.1.63	192.168.1.1 – 192.168.1.62	**62**
/27	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94	**30**

/28	192.168.1.96	192.168.1.111	192.168.1.97 – 192.168.1.110	**14**

Step 2: Usable and Unused IP Addresses

Subnet	Total Addresses	Usable Hosts	Reserved (Network + Broadcast)
/26	64	62	2
/27	32	30	2
/28	16	14	2

Example (assuming departments need 20, 15, and 10 hosts):

Department	Allocated	Needed	Unused

Department A (/26)	62	20	**42**
Department B (/27)	30	15	**15**
Department C (/28)	14	10	**4**

Step 3: Identify Inefficient Subnet Allocations

- Department A has **42 unused IP addresses**, so **/26 is oversized**.
- Department B has **15 unused IP addresses**, so **/27 may be larger than required**.
- Department C has only **4 unused IP addresses**, making **/28 a better-sized subnet**.

Step 4: Debug the Addressing Plan

Possible mistakes:

- Departments with few users assigned **large subnets**.
- Departments requiring many hosts assigned **small subnets**, causing insufficient addresses.
- Subnet overlap due to incorrect subnet boundaries.
- Wasted address space because fixed-size subnetting was used instead of VLSM.

Step 5: Optimized VLSM Scheme

Assign the largest subnet first, then smaller ones.

Depar	**Hosts**	**VLSM**	**Network**	**Host Range**

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Department A	50	/26	192.168.1.0/26	192.168.1.1 – 192.168.1.62
Department B	20	/27	192.168.1.64/27	192.168.1.65 – 192.168.1.94
Department C	10	/28	192.168.1.96/28	192.168.1.97 – 192.168.1.110

This VLSM allocation:

- Prevents overlapping subnets.
- Reduces wasted IP addresses.
- Supports current departmental requirements.
- Leaves the remaining addresses (192.168.1.112–192.168.1.255) available for future expansion.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.
<u>Task</u>

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

SOL:**IPv6 Configuration Debugging**

Step 1: Expand the IPv6 Addresses

Device	Compressed Address	Expanded Address
Device A	2001:db8::ff00:42:8329/64	**2001:0db8:0000:0000:0000:ff00:0042:8329/64**
Device B	2001:db8:0:0:1::1/64	**2001:0db8:0000:0000:0001:0000:0000:0001/64**

Step 2: Determine the Network Prefix

For a **/64** prefix, the **first four blocks (64 bits)** represent the network.

Device	Network Prefix
Device A	2001:db8:0:0::/64
Device B	2001:db8:0:0::/64

Step 3: Verify Whether Both Devices Belong to the Same Subnet

- Device A Prefix: **2001:db8:0:0::/64**
- Device B Prefix: **2001:db8:0:0::/64**

Result: ✅ Yes, both devices belong to the same subnet.

Step 4: Identify the Addressing or Prefix Mismatch

- There is **no IPv6 address or prefix mismatch**.
- Both devices are configured in the **same /64 subnet**.
- If communication fails, the problem is likely due to:
  - Incorrect default gateway

- Interface/link down
- Firewall settings
- Duplicate IPv6 address
- Neighbor Discovery (NDP) issue

Step 5: Optimized IPv6 Configuration

Assign unique addresses within the same subnet:

Device	Optimized IPv6 Address
Device A	**2001:db8:0:0::10/64**
Device B	**2001:db8:0:0::20/64**

Both devices remain in the **2001:db8:0:0::/64** network.

Step 6: Available Host Addresses

- Prefix Length: **/64**
- Host Bits: **64**

Available Host Addresses = 2^{64}

2^{64} = **18,446,744,073,709,551,616** host addresses.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

SOL:**IPv6 Configuration Debugging**

Step 1: Expand the IPv6 Addresses

Device	**Compressed Address**	**Expanded Address**
Devic	2001:db8::ff00:42:8329/64	**2001:0db8:0000:0000:0000:ff00:0042:8329/64**

e A		
Device B	2001:db8:0:0:1::1/64	**2001:0db8:0000:0000:0001:0000:0000:0001/64**

Step 2: Determine the Network Prefix

For a **/64** prefix, the **first four blocks (64 bits)** represent the network.

Device	**Network Prefix**
Device A	**2001:db8:0:0::/64**
Device B	**2001:db8:0:0::/64**

Step 3: Verify Whether Both Devices Belong to the Same Subnet

- Device A Prefix: **2001:db8:0:0::/64**
- Device B Prefix: **2001:db8:0:0::/64**

Result: ✅ Yes, both devices belong to the same subnet.

Step 4: Identify the Addressing or Prefix Mismatch

- There is **no IPv6 address or prefix mismatch**.
- Both devices are configured in the **same /64 subnet**.
- If communication fails, the problem is likely due to:
  - Incorrect default gateway
  - Interface/link down
  - Firewall settings
  - Duplicate IPv6 address
  - Neighbor Discovery (NDP) issue

Step 5: Optimized IPv6 Configuration

Assign unique addresses within the same subnet:

Device	Optimized IPv6 Address
Device A	**2001:db8:0:0::10/64**
Dev	**2001:db8:0:0::20/64**

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B

Both devices remain in the **2001:db8:0:0::/64** network.

Step 6: Available Host Addresses

- Prefix Length: **/64**
- Host Bits: **64**

Available Host Addresses = 2^{64}

2^{64} = 18,446,744,073,709,551,616 host addresses.

Final Answer

Item	Answer
Expanded Address (Device A)	**2001:0db8:0000:0000:0000:ff00:0042:8329/64**
Expanded Address (Device B)	**2001:0db8:0000:0000:0001:0000:0000:0001/64**
Network Prefix (Both)	**2001:db8:0:0::/64**
Same Subnet?	**Yes**
Addressing/Prefix Mismatch	**None**

Optimized Addresses	Device A: **2001:db8:0:0::10/64**, Device B: **2001:db8:0:0::20/64**
Available Host Addresses	**2^{64} = 18,446,744,073,709,551,616**

A router contains the following routing table. Destination Network Next Hop 192.168.1.0/24 10.0.0.2 192.168.2.0/24 10.0.0.3 Default Route 10.0.0.1 Users report that packets destined for 192.168.2.100 are not reaching the intended network. Task Debug the routing configuration by: Identifying the matching routing entry using the longest-prefix match. Determining the next-hop router selected for packet forwarding. Verifying whether the destination IP belongs to the intended subnet. Identifying any routing configuration errors. Optimizing the routing table to improve packet forwarding efficiency. Identifying the route that will be used if no matching destination exists.

Routing Configuration Debugging

Step 1: Identify the Matching Routing Entry (Longest-Prefix Match)

Destination IP: 192.168.2.100

Routing Table:

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2

192.168.2.0/24	**10.0.0.3**
Default Route (0.0.0.0/0)	10.0.0.1

Matching Route: 192.168.2.0/24

Step 2: Determine the Next-Hop Router

- Destination IP: **192.168.2.100**
- Matching Route: **192.168.2.0/24**
- **Next-Hop Router: 10.0.0.3**

Step 3: Verify the Destination Subnet

- Network Address: **192.168.2.0/24**
- Host Range: **192.168.2.1 – 192.168.2.254**
- Broadcast Address: **192.168.2.255**

Since **192.168.2.100** falls within **192.168.2.1–192.168.2.254**:

Result: ✅ The destination belongs to the **192.168.2.0/24** subnet.

Step 4: Identify Routing Configuration Errors

The routing table appears **correct**.

If packets are not reaching the destination, possible issues include:

- Next-hop router **10.0.0.3** is unreachable.
- Incorrect subnet mask on the destination network.
- Missing return route.

- Interface is down.
- ACL or firewall blocking traffic.

Step 5: Optimize the Routing Table

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route (0.0.0.0/0)	10.0.0.1

Optimization:

- Verify that **10.0.0.3** is reachable.
- Remove any duplicate or unnecessary routes.
- Keep specific routes before the default route for efficient forwarding.

Step 6: Route Used if No Match Exists

If no destination network matches, the router uses the:

Default Route (0.0.0.0/0) → Next Hop: 10.0.0.1

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues

Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation